FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Circular Laminae

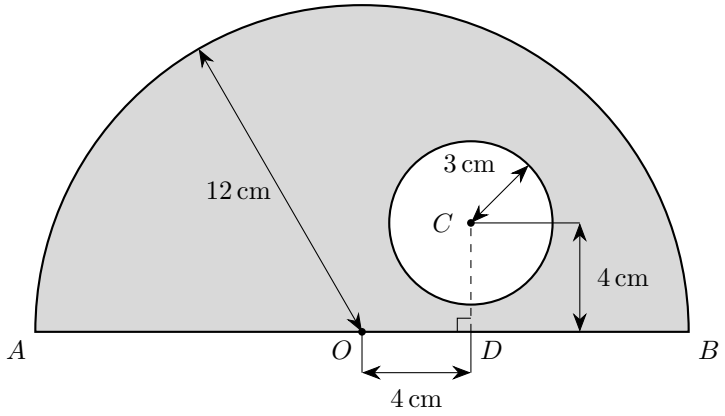
Questions by topic 7 questions · 76 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*9 marks*	2
Question 2	*9 marks*	3
Question 3	*8 marks*	4
Question 4	*9 marks*	5
Question 5	*12 marks*	6
Question 6	*14 marks*	7
Question 7	*15 marks*	8

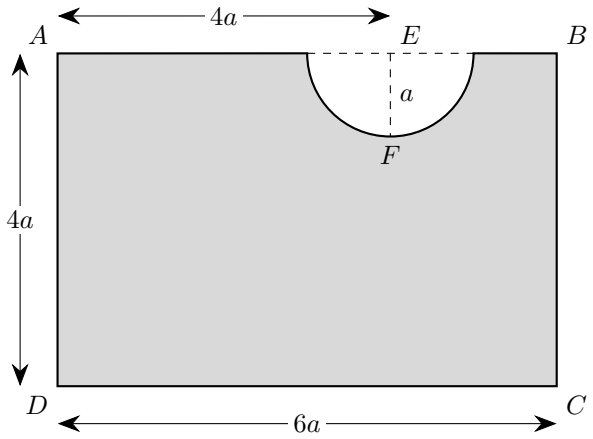


1. A uniform semicircular lamina has radius 12 cm and diameter AB, with midpoint O.
A circular hole of radius 3 cm is cut from the lamina.
Its centre C is 4 cm above a point D on OB, where $OD = 4$ cm and CD is perpendicular to AB, as shown in the diagram above.

The remaining lamina is freely suspended from B and hangs in equilibrium.

Determine the acute angle that AB makes with the vertical, giving your answer to the nearest $0.1°$.

You may use the result that the centre of mass of a uniform semicircular lamina of radius R is on its axis of symmetry at a distance $\frac{4R}{3\pi}$ from its diameter. **[9]**



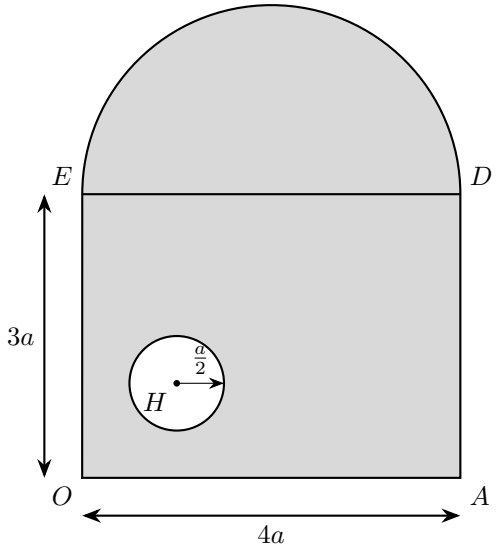
2. A uniform rectangular lamina $ABCD$ has $AB = 6a$ and $AD = 4a$, where $a > 0$.
A semicircular notch of radius a is cut from the edge AB.
The centre E of the semicircle lies on AB, with $AE = 4a$, and the semicircle lies inside the rectangle, as shown in the diagram above.

You may use the result that the centre of mass of a uniform semicircular lamina of radius r is a distance $\frac{4r}{3\pi}$ from its diameter.

(a) (i) Show that the perpendicular distance of the centre of mass of the remaining lamina from AD is

$$\frac{4(36-\pi)}{48-\pi}a$$

(ii) The remaining lamina is freely suspended from A and hangs in equilibrium. Calculate the acute angle that AB makes with the downward vertical, giving your answer to the nearest $0.1°$. **[9]**



3. A uniform lamina is made by joining a rectangle $OADE$ to a semicircle along its diameter ED, and then removing a circular disc from the rectangular portion, as shown in the diagram above.

The rectangle has $OA = 4a$ and $OE = 3a$, where $a > 0$.
The removed disc has radius $\frac{a}{2}$ and centre H, which is a distance a from each of the sides OE and OA.

You may use the result that the centre of mass of a uniform semicircular lamina of radius r is a distance $\frac{4r}{3\pi}$ from its diameter.

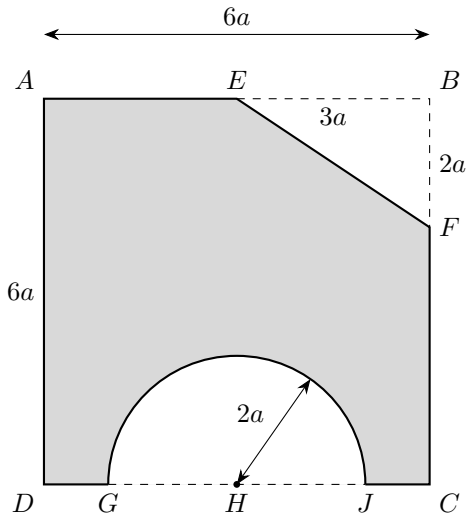
(a) Find, in terms of a and π, the distance of the centre of mass of the remaining lamina from

(i) OE,

(ii) OA. **[6]**

The remaining lamina is freely suspended from A and hangs in equilibrium.
The angle between AO and the downward vertical is α.

(b) Find the exact value of $\tan\alpha$. **[2]**



4. A uniform square lamina $ABCD$ has side $6a$.
A template T, shown shaded in the diagram above, is made by removing two portions of the lamina.

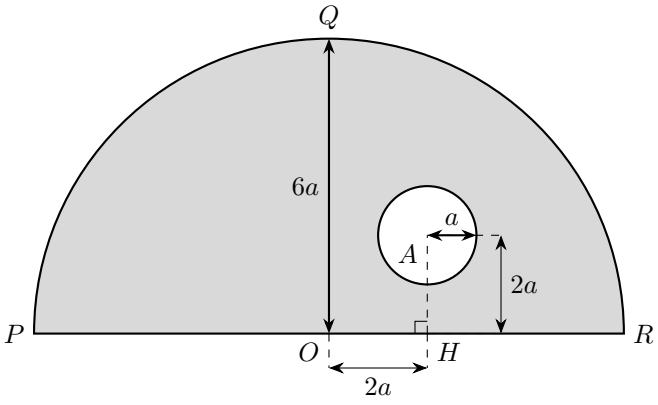
The first portion is the right-angled triangle EBF, where E lies on AB, F lies on BC, $EB = 3a$ and $BF = 2a$.

The second portion is a semicircle of radius $2a$.
Its centre H is the midpoint of DC, its diameter lies on DC, and the semicircle lies inside the square.

(a) Find the distances of the centre of mass of T from AD and from DC, giving both answers exactly in terms of a and π. **[6]**

The template is now suspended freely from A and hangs in equilibrium in a vertical plane.

(b) Find the acute angle between AD and the vertical, giving your answer in degrees to one decimal place. **[3]**



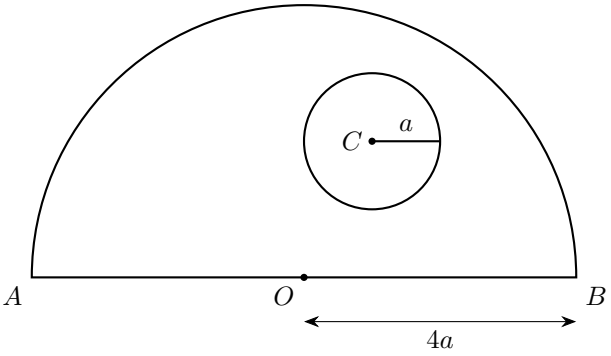
5. A uniform semicircular lamina has centre O, radius $6a$ and diameter PR.
A circular disc of radius a is removed to form the lamina L shown shaded in the diagram above.
The centre of the removed disc is A.
The point H lies on OR, with $OH = 2a$, $HA = 2a$ and HA perpendicular to PR.

Take O as the origin, with the positive x-axis directed towards R and the positive y-axis directed towards Q.

(a) Find the exact coordinates of the centre of mass of L. **[6]**

The mass of L is M.
A particle of mass kM is attached at P.
The system is suspended from R and hangs freely in equilibrium, with P below R and the diameter PR making an acute angle of $\arctan\left(\frac{1}{6}\right)$ with the downward vertical through R.

(b) Find the exact value of k and the magnitude of the supporting force at R in terms of M and g. **[6]**

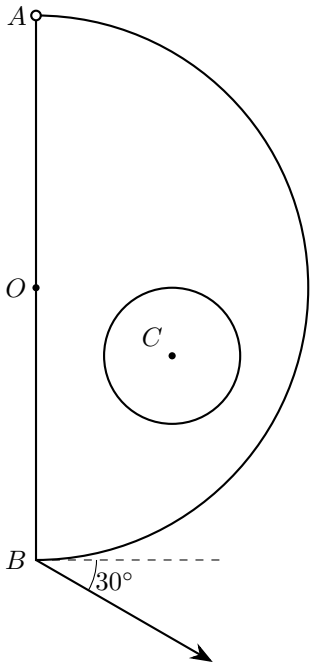


6. A uniform semicircular lamina has radius $4a$ and diameter AB, whose midpoint is O.
A circular hole of radius a is cut from the lamina.
In the orientation shown above, take O as the origin, the positive x-direction along OB and the positive y-direction perpendicular to AB towards the semicircular arc.
The centre C of the hole has coordinates $(a, 2a)$.

(i) Show that the centre of mass G of the remaining lamina has coordinates

$$\left(-\frac{a}{7},\ \frac{(128-6\pi)a}{21\pi}\right)$$ **[6]**

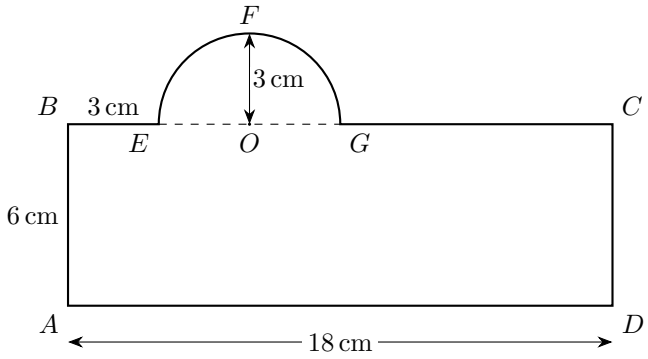
The remaining lamina has mass 7 kg.
It is freely pivoted at A and turned so that AB is vertical, with A above B and the semicircular arc to the right of AB, as shown in the diagram below.
A light string attached to B holds the lamina in equilibrium.
The string lies in the plane of the lamina and makes an angle of 30° below the horizontal.



(ii) Calculate the tension in the string. **[3]**

(iii) Find the magnitude of the force exerted on the lamina by the pivot at A and the angle this force makes with the upward vertical, stating on which side of the vertical it acts. **[5]**

7. A uniform lamina consists of a rectangle $ABCD$ together with a semicircular tab EFG attached externally along its diameter EG, as shown in the diagram below. The two components have the same mass per unit area. The points E and G lie on BC, and O is the midpoint of EG.
The dimensions are $AB = 6\,\text{cm}$, $BC = 18\,\text{cm}$ and $BE = 3\,\text{cm}$. The semicircular tab has radius $3\,\text{cm}$.



(i) Consider the semicircular tab EFG on its own. Using the standard centre-of-mass result for a uniform semicircular lamina, show that its centre of mass is $1.27\,\text{cm}$ from the diameter EG, correct to 3 significant figures. State the exact distance. **[2]**

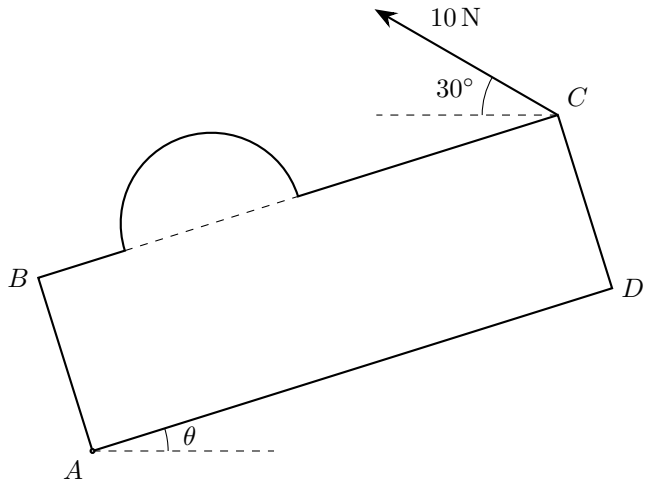
(ii) For the complete lamina, find

(a) the distance from AB to its centre of mass, giving your answer exactly, **[4]**

(b) the distance from AD to its centre of mass, giving your answer exactly. **[3]**

(iii) The complete lamina has weight $24\,\text{N}$ and is smoothly pivoted at A. A wire attached at C supports it in equilibrium. The wire pulls upwards and to the left at $30°$ above the horizontal, and its tension is $10\,\text{N}$.

The side AD makes an angle θ above the horizontal, where $0° < \theta < 45°$, as shown in the diagram below.



Calculate θ, giving your answer to 3 significant figures. **[6]**